

SiteSentinel

*AI-Powered Construction Site Safety Monitoring
& Smart Attendance System*

Theme: Smart Safety & Automation

Sub Theme: Construction Site Worker Safety & Workforce Management

Title: SiteSentinel — AI-Powered PPE Monitoring & Smart Attendance System

1. Original Prototype & New Feature Being Added

Original Prototype

SiteSentinel is an AI, Computer Vision, and IoT-integrated construction site safety monitoring platform. The original prototype was built to solve two critical real-world problems at construction sites: ensuring workers wear proper Personal Protective Equipment (PPE), and automating worker attendance without any manual intervention.

Core Components of the Original Prototype:

- **PPE Detection Engine** — A custom-trained YOLOv8 model detects PPE items (hardhat, safety vest, mask) and violations (NO-Hardhat, NO-Safety Vest, NO-Mask) in real time from a live camera stream with 10 detection classes.
- **Face Recognition Attendance** — Workers are identified using face_recognition (dlib-based) by comparing live face embeddings against registered worker embeddings stored in the cloud database, enabling automatic check-in/out.
- **Raspberry Pi IoT Integration** — A Raspberry Pi 4 acts as the remote camera node and IoT gateway — streaming live MJPEG video and hosting HTTP endpoints to control a buzzer and LED Matrix display.
- **Flask Backend & Turso Database** — A Flask-based REST API backend stores worker profiles, face embeddings, attendance logs, and PPE scores in a Turso (LibSQL) cloud database accessible from anywhere.
- **Manager Dashboard** — A web dashboard (HTML/CSS/JS + Tailwind) allows site managers to register workers, upload face images, and monitor real-time attendance and PPE compliance history.

New Feature Currently Being Added:

The major new addition is the **Per-Worker PPE Compliance Scoring System** with real-time IoT feedback — bridging AI detection directly to physical on-site alerts:

- **PPE Score Calculation** — A numerical PPE score (0–100) is computed per worker at every check-in and check-out event, based on the detected safety equipment at the exact moment of face recognition.

- **Score-to-Attendance Binding** — The score is linked directly to the worker's attendance record in the database, creating a personal, auditable safety compliance history over time.
- **LED Matrix Live Feedback** — The score is pushed via HTTP to the Raspberry Pi, which scrolls it on the LED Matrix in real time — giving immediate on-site visual feedback to both the worker and the supervisor.
- **Buzzer Violation Alert** — The buzzer triggers an audible alert when a PPE violation is detected, warning the worker on-site instantly without requiring any manual intervention.
- **ESP32 Multi-Zone Expansion** — ESP32 microcontrollers (firmware already drafted) are being integrated as lower-cost IoT nodes for wider site coverage, supplementing the Raspberry Pi.

2. How This Update Improves Upon Existing Solutions

Construction site safety monitoring is traditionally handled through manual inspections, generic CCTV footage review, or expensive enterprise platforms — all of which are reactive, slow, and costly. SiteSentinel addresses these gaps through a smarter, unified, and affordable approach.

Comparison with Existing Market Solutions:

Aspect	Traditional / Market Solutions	SiteSentinel's Approach
Attendance	Manual registers or RFID cards	Automatic face recognition check-in/out
PPE Monitoring	Manual inspection by safety officers	Real-time AI detection — YOLOv8 (10 classes)
Worker Identity	Badge scanning — shared badges cause errors	Biometric face recognition — unique per person
Alert Speed	Post-incident reports — hours or days later	Instant buzzer + LED Matrix alert in seconds
Cost	Enterprise systems cost lakhs per deployment	Open-source stack on Raspberry Pi — minimal cost
Data Integration	Siloed: camera, attendance, and safety are separate	Unified: camera → AI → attendance + PPE → database
Accountability	No link between attendance and safety compliance	PPE score tied to each worker's attendance record
Scalability	Fixed CCTV, expensive to expand across zones	ESP32 nodes for low-cost multi-zone coverage

Key Innovation Points:

- **Identity-Linked Safety Accountability** — SiteSentinel uniquely ties worker biometric identity to real-time PPE compliance, creating an accountable personal safety record per worker — not just a generic site-level alert.
- **Affordable Edge AI** — Runs on commodity hardware (Raspberry Pi + standard GPU laptop), deployable at a fraction of the cost of enterprise safety platforms.
- **Physical Feedback Loop** — Buzzer and LED matrix alerts bridge the gap between AI detection on a PC and the actual worker on the ground, where real-time response matters most.

- **GPU-Accelerated Real-Time Inference** — PyTorch CUDA (NVIDIA RTX 4050) enables smooth real-time detection without dedicated AI hardware, keeping the solution cost-effective and accessible.
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3. Biggest Insight, Unexpected Hurdle & Next Steps

Biggest Unexpected Hurdle:

The most critical challenge was **preventing false PPE score generation**. In early testing, the system occasionally recorded PPE violation scores from background objects — such as a hardhat lying on a table or a safety vest hanging on a wall — without any worker actually being present in frame, leading to incorrect compliance records in the database.

The fix required building a strict two-condition validation gate: a PPE score is computed and logged ONLY when (a) a face is successfully recognized with sufficient confidence, AND (b) a 'Person' class bounding box is simultaneously detected by YOLOv8 in the same frame. This dual-condition check eliminated nearly all false positives and made PPE scoring reliable.

Biggest Insight:

The most important architectural discovery was the need to **split computation across devices**. Initially, the team attempted to run AI inference directly on the Raspberry Pi — which was far too slow for real-time use. The breakthrough was re-architecting so the Pi handles only camera streaming and IoT actuation (buzzer, LED), while all heavy AI work (YOLOv8 + face recognition) runs on the main PC with GPU. This split made the system practically viable at very low hardware cost.

Immediate Next Steps to Make SiteSentinel Fully Production-Ready:

- **1. Refine PPE Scoring Logic** — Handle edge cases such as partial face occlusion (worker wearing a mask), multiple workers in a single frame, and overlapping bounding boxes to improve accuracy.
 - **2. Violation Alert Log on Dashboard** — Add a dedicated log with timestamps, worker ID, and captured frame snapshots for safety audit and compliance documentation.
 - **3. Harden Face Recognition in Poor Lighting** — Calibrate the recognition confidence threshold for varying lighting conditions to reduce false positives in low-light or high-glare outdoor environments.
 - **4. Cloud Deployment of Backend** — Deploy the Flask backend to a cloud server (Render config already created in the repo) for remote access by site managers from any device.
 - **5. ESP32 Multi-Zone IoT Expansion** — Integrate ESP32 nodes (firmware already drafted) as lightweight, low-cost IoT devices for multi-zone site coverage beyond the Raspberry Pi's single-point range.
 - **6. Mobile-Friendly Manager Dashboard** — Build a responsive mobile version of the dashboard for real-time monitoring from smartphones on-site, without needing a laptop.
 - **7. Automated Daily Safety Reports** — Generate automated PDF reports summarising per-worker attendance, PPE scores, and violations each day for management review and regulatory compliance.
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